

PELLA, IA, USA

WMO: 720312

Lat: 41.400N

Lon: 92.946W

Elev: 885

StdP: 14.23

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 04964

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-3.9	0.7	-10.5	3.2	-2.3	-6.8	4.0	2.0	28.3	20.6	25.9	20.9	6.8	320	0.496

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.1	93.0	75.8	90.4	74.3	87.1	72.9	78.4	87.5	76.7	85.6	75.0	84.0	8.3	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
76.2	141.6	84.1	73.5	128.7	81.4	72.4	123.9	80.2	42.6	87.5	40.7	85.3	39.0	83.7	84.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.7	20.2	18.0	-12.0	97.3	5.7	3.5	-16.1	99.8	-19.5	101.9	-22.7	103.9	-26.9	106.4
			-12.9	81.7	5.7	2.1	-17.0	83.2	-20.4	84.5	-23.6	85.6	-27.8	87.1
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	51.3	23.7	26.1	39.4	51.6	62.7	72.0	75.6	72.9	66.9	53.5	40.2	29.1		
	DBStd	20.59	12.58	12.34	11.15	9.90	8.55	5.61	5.71	5.29	7.78	9.46	10.46	11.31		
	HDD50	3000	817	671	362	97	9	0	0	0	1	70	322	651		
	HDD65	6038	1282	1088	794	411	146	9	2	4	70	374	744	1114		
	CDD50	3464	0	3	34	146	403	659	794	709	507	180	28	1		
	CDD65	1029	0	0	2	10	76	218	331	248	126	18	0	0		
	CDH74	9618	0	0	16	141	670	1989	3346	2150	1132	172	2	0		
	CDH80	3373	0	0	1	23	171	668	1359	737	379	35	0	0		
(14) Wind	WSAvg	7.6	9.2	8.8	9.2	9.5	8.1	6.5	5.1	4.5	5.7	7.4	8.8	8.6		
(15) Precipitation	PrecAvg	36.4	1.0	1.4	2.2	3.7	4.7	5.5	4.5	4.5	3.5	2.5	1.9	1.2		
	PrecMax	52.9	2.3	4.3	5.0	8.2	10.7	13.1	11.7	13.1	8.9	5.8	4.5	4.3		
	PrecMin	24.6	0.6	0.2	0.1	1.0	0.8	0.8	0.7	1.0	0.9	0.6	0.0	0.0		
	PrecStd	8.0	0.4	0.9	1.4	1.8	2.2	2.8	3.0	3.1	1.8	1.7	1.2	1.0		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.3	64.7	76.4	83.6	90.4	93.2	99.7	95.4	92.2	85.0	72.8	59.2		
		MCWB	45.4	52.4	62.2	63.9	71.1	76.5	75.3	72.6	73.3	68.1	58.2	52.1		
	2%	DB	48.4	55.2	68.8	78.7	84.2	90.6	94.4	91.1	88.4	79.0	65.6	52.6		
		MCWB	41.8	46.1	56.5	61.1	67.9	73.8	76.9	74.6	72.4	64.5	55.4	46.9		
	5%	DB	44.1	48.0	62.6	73.2	81.1	87.0	90.8	87.3	84.3	73.4	60.9	48.4		
		MCWB	39.3	41.8	52.6	58.2	65.4	71.8	75.6	73.1	70.3	62.3	52.3	43.9		
	10%	DB	39.3	42.7	56.5	68.7	77.5	83.5	87.1	83.7	80.9	69.6	56.1	43.5		
		MCWB	36.0	38.4	48.9	55.6	64.5	70.7	74.5	71.7	67.4	59.4	48.5	39.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	47.2	54.1	62.4	66.7	73.4	79.8	81.4	80.4	77.7	71.8	61.1	55.5		
		MCDB	51.7	63.1	73.5	79.8	84.2	88.4	90.4	90.0	87.1	79.6	67.4	58.5		
	2%	WB	43.1	47.6	58.6	63.2	70.9	76.8	78.9	77.6	74.7	68.3	57.7	49.1		
		MCDB	48.5	55.7	66.2	74.1	80.7	85.8	88.5	85.3	84.0	75.4	63.9	51.4		
	5%	WB	39.4	42.5	53.5	60.3	68.7	74.6	77.2	75.4	72.4	65.0	53.5	43.6		
		MCDB	43.5	47.7	61.4	69.9	78.1	83.6	86.6	82.9	80.8	70.5	59.6	47.6		
	10%	WB	36.2	38.2	49.0	57.7	66.7	72.8	75.5	73.4	70.4	60.5	49.5	39.6		
		MCDB	39.4	42.6	56.6	66.9	74.9	81.2	85.0	81.7	77.7	67.2	55.4	43.2		
(35) Mean Daily Temperature Range	5% DB	MDBR	15.8	17.0	18.2	21.0	19.9	19.6	20.1	20.1	21.8	20.1	18.0	15.1		
		MCDBR	20.8	24.3	28.1	28.2	23.3	22.8	23.0	23.6	24.3	25.7	24.8	20.8		
	5% WB	MCWBR	15.5	16.4	16.7	15.1	11.1	10.7	9.3	10.2	10.3	13.7	15.8	15.7		
		MCWBR	20.2	23.7	23.7	23.7	20.2	20.0	19.5	19.4	20.5	19.9	22.9	19.1		
(40) Clear-Sky Solar Irradiance	taub		0.288	0.311	0.333	0.385	0.411	0.421	0.438	0.419	0.385	0.341	0.300	0.289		
		taud	2.422	2.361	2.390	2.262	2.261	2.278	2.219	2.290	2.335	2.455	2.499	2.437		
	Ebn at Noon		276	285	291	281	275	272	265	267	269	271	270	265		
		Edh at Noon	25	31	34	41	43	42	44	40	36	28	23	23		
(44) All-Sky Solar Radiation	RadAvg		608	927	1203	1505	1770	1963	1980	1723	1463	978	663	508		
	RadStd		58	65	111	111	134	196	146	105	108	112	57	48		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (46 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page